

Artificial Intelligence (CT653 / CT710 / CT78506)

Concise PYQ, 2069–2082

Notes: TU IOE PYQ 2069–2082, 41 papers. Regular = **bold**, Back = normal. Three course codes: CT653 (BCT, III/II) = normal, CT710 (BEI, IV/I) = **mono**, CT78506 (“AI Elective III”, BEX, IV/II) = *italic*. Marks in (), unique values only. Keys: (I)=Importance/Need, (+)=Advantage, (-)=Disadvantage, +Fig=Figure, +Eg=Example, Vs=Compare/Differentiate.

1. Introduction

- Define AI; AI Vs Natural (human) intelligence; can AI replace humans; subfields; think/act rationally (7)(6)(1+3+4)(8)(3+5)(4)
- Turing test (I); measure machine intelligence; total Turing test; Reverse Turing test; can machine think; passed $\Rightarrow$ intelligent; factors required to pass it; +applications; +(-) of AI (7)(2+5)(4+3)(2+4)(2+4+2)(2+3+3)(2+2+4)
- Intelligent / rational agent; types +Eg; agent-environment; learning agent (1+6)(2+6)(8)
- PEAS — medical diagnosis, taxi driver, factory robot, self-driving car, medicine-delivery drone (2+3)(1+5)(2+2+3)(4+4)
- History of AI +chronological; “AI research failure” period (2+5)(2+6)(7)
- Applications of AI / fields in daily life / AI as problem-solving tool; growth areas; (-) of AI (7)(6)(2+4)
- Knowledge Vs learning; acting humanly; belief Vs hypothesis Vs knowledge (2+5+1)(4+4)(4)
 - Learning essential for agents +role; +Supervised Vs Unsupervised (4+4)(2+2+2+4)

2. Problem Solving

- Well-defined problem +characteristics; problem space; state, successor fn, goal test, path cost; problem-solving steps; define a problem in state space (3+4)(8)(2)(3)
- Water jug — 3/5/9 gal (get 7), 4/3 gal (get 2), 5/2 litre (get 1) +production rules +search tree (2+1+2+3)(2+2+4)(4)(8)
- Farmer goat/wolf/cabbage — formulate + solve +tree (8)
- CSP +constraints +two examples; crypto-arithmetic (7)(1+6)(2+6)(3+5)(5+3)(8)(1)(3)(2+5)(3+6)(3+4)
 - LOGIC+LOGIC=PROLOG; RIGHT+RIGHT=WRONG
 - WRONG+WRONG=RIGHT; ONE+ONE+TWO=FOUR
 - TWO+TWO=FOUR; SEND+MORE=MONEY
 - SWIM+WEAR=RELAX; BASE+BALL=GAMES
 - LOVE+LOVE=HATE; EAT+THAT=APPLE
 - TEN+TEN+FORTY=SIXTY; FORTY+TEN+TEN=SIXTY
 - CROSS+ROADS=DANGER; AB+CD=AAA (values of B)
- Production system +essential steps +Eg; why searching needed (2+6)(4)(2)
- Game playing — Tic-Tac-Toe state space, data structures, goal (7)

3. Search Techniques

- Necessity of searching +justify; types of search methods; evaluation criteria for search algorithms (2+7)(4+4)(2)(3)
- BFS Vs DFS +performance; why blind; depth-limit search (+)(9)(4+4)(4+5)(1+7)(7)(8)
- Informed Vs uninformed / blind search +Eg (8)(4+4)(3+6)
- Hill climbing +problems +solutions; simulated annealing; local maxima (9)(6+2)(3+5)(4+4)
- A* path+heuristic +Eg; greedy drawbacks Vs A*; admissibility & optimality (8)(2+7)(2+6)(4+5)
 - +Fig S $\rightarrow$ G cost graph, heuristic table — compare A* Vs greedy (8)(3+4+3)
 - Best First Search — open list / closed list / path at each step +Fig (5)
- Minimax (-) +alpha-beta pruning +Fig(tree) (2+6)(2+4+3)(4+4)(6)
- Short notes: A* search; uninformed search; adversarial search (3)(4)

4. Knowledge Rep., Inference & Reasoning

- CNF need + conversion steps +Eg; FOL $\rightarrow$ CNF steps; CNF + rules of inference (10)(8)(2+6)(5)(2+3)(2+2+2+3)(2+2+4)
 - Transform: $P \vee (\neg P \wedge Q \wedge R)$; $P \rightarrow ((Q \wedge \neg R) \rightarrow S)$; $P \rightarrow ((Q \wedge \neg R) \leftrightarrow S)$;
 $\neg(P \wedge Q) \vee (P \vee S) \rightarrow R$
 - “Everyone who loves all animals is loved by someone” $\rightarrow$ FOPL $\rightarrow$ CNF
- FOPL need + rules of inference (when to use); horn clause; skolemization +Eg; WFF +tautology; FOL conversion; predicate logic; sentence validity (2+3+3)(2)(4+4)(3)(4)
- Resolution refutation proofs (6)(7)(8)(9)(10)(2+8)(3+3)(6+4)(3+5)(2+6)
 - American/weapons-criminal (John, Donald, Col. West, George, Mr. A); oversmart-Ram; block-world supports; pompeian-Caesar

- likes-food chains: John/Ram/Sita/Bhogendra-peanuts, mushroom, chicken; Charlie-horse/mammal; Kali/Taarey-cow
- Mary/John-study; Sneha-happy; Santa/Rudolph-Scrooge; “Did Curiosity kill the cat”; hounds/direwolves-light sleeper-mice; Steve-BK301; exciting life
- Forward Vs backward chaining +Eg; rule-based reasoning; knowledge-based agent (4+4)(7)(2+5)(2+3+2)(2+4)
- Bayes / causal / belief network reasoning +Eg; prior Vs posterior; uncertainty measures; why probabilistic reasoning matters (2+5)(2+6)(7)(3+5)(8)(5+3)(3+6)
 - meningitis/stiff-neck (50%, 80%); flu/measles rash; county 51%/cigar; mammogram; genetic defect 1%/90%/10%; 3 factories/defective bulbs; university 6-feet/60% female; rare disease 99% test
 - BBN +Fig: cloudy/sprinkler/rain/wet grass; smoker $\rightarrow$ cancer $\rightarrow$ test

5. Structured Knowledge Representation

- KR models / approaches / issues / properties; how KR evaluated + choosing among approaches; 4 KR schemes (7)(2+5)(3+5)(2+6)(4+2+2)(8)(2)(2+2+2+2)
- Semantic net +Eg; when to use one; convert sentences $\rightarrow$ net; +learning by analogy (8)(7)(2+6)(1+6)(1+7)
 - Tom the cat (Ram / Sudeep); Pee-wee-Reese / Rohit Paudel batting average; Sakti Gauchan / Sandeep Lamichane; Tweety & Sweetie; Figo the dog; employee/supervisor; cat-mammal-fish; Ram-Jhilke
- Define frame; frames +Eg +significance; frame Vs semantic net (+)(-); convert net $\rightarrow$ frame (7)(4+4)(3+5)(6+2)(8)(1)(4+3)(5)
- Conceptual dependency + primitives; scripts (2+5)(3+5)

6. Machine Learning

- Define ML; methods / ways of learning; learning framework +Fig +Eg; supervised Vs unsupervised Vs reinforcement; clustering; learning essential for agents (3+5)(2+6)(2+4)(8)(1)(4+4)(5)(6)
- Learning by analogy; inductive learning / ID3 +Eg; induction Vs deduction (1+7)(2+6)(3)
 - Decision tree root attribute — Play golf table; Play cricket table; mushroom (edible) table (8)(1+7)(2+7)
- Genetic algorithm — steps / operators +Fig +Eg; selection, crossover, mutation (I); evolutionary computing; why the study of genes matters to AI; applications (10)(2+8)(8)(2+6)(2+6+2)(3+5)(7)(4+4)(5)
 - Swarm intelligence; fitness $f(x)=(a+b)-(c+d)+(e+f)-(g+h)$ (2+8)
- Fuzzy logic Vs binary logic; when to use it; fuzzy learning / inference (Mamdani) +Fig +Eg; membership; FIS architecture; steps to develop a fuzzy system (4)(2+6)(3+7)(8)(2+5)(1+2+5)
- Boltzmann machine +algo (1+7)(2+8)

7. Applications of AI

Neural Networks

- Backprop algo; math model + train NN; ANN non-linearity; activation function role (4+4)(2+4+4)(3+5)(2+2+4)(1+7)
- McCulloch-Pitts model +Fig; AND (not XOR); choosing activation function (8)(10)(3+4)(4+3)
- Perceptron / MLP as AND, OR, XOR gate (or logic gates in general); linear classifier; perceptron algorithm; MLP working principle (2+7)(1+7)(2+6)(1+8)(6)(1+6)(7)
- Forward propagation numerical +Fig (weights a-f, ReLU then sigmoid, update a, b) (8)
- Hebbian network (AND) +is it supervised (3+7)(6+3)(1+7)
- Hopfield — steps +Eg, features, active/passive units, store 3 patterns of length 4, Vs feed-forward (8)(6)(3+5)
- SOM / Kohonen +steps +Eg; Hopfield Vs Kohonen; learning rate (2+5)(3)(4)(5)

Expert System

- Development steps +Fig block; scenario (disease diagnosis, disaster) (8)(4+4)(2+6)(6)
- Architecture +components +block diagram; characteristics; inference; ES shells; +(+)(-) and benefits (2+4)(2+5)(3+5)(1+7)(2+2+3)(1+4+3)(5+2+3)
- Declarative Vs procedural knowledge + architecture (2+6)(3+6)(6)
- (+)(-) and limitations of ES; ES Vs human expert; knowledge acquisition +Eg (8)(4+3)(2+6)(2+5)

NLP & Machine Vision

- NLP steps +Fig +Eg; NLU/NLG; why NLP is hard; +major issues
(6)(8)(2+4)(3+5)(2+7)(5+2)(9)(1+6)(7)(6+2+2)
- Machine translation +challenges; NLP issues / ambiguity
(1+7)(2+6)(2+5)(2+4+2)

- Levels of analysis — phonetic / syntactic / semantic / pragmatic / morphological; parse tree
(2+2+2+2)(8)(5+4)
- Nepali NLP — challenges +mitigation; Nepali chatbot +Fig (8)(2+6)
- Machine / computer vision (I) +steps +Eg +applications
(3+5)(2+5)(1+5+2)